AI-Based Classroom Learning Companion

Final Year Project

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A project submitted in partial fulfillment of the degree of

BS in Computer Science

Department of Computer Science

Faculty of Computer Science & Information Technology

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Type (Nature of project)	In our project developing an AI-based classroom learning companion, our key focus is on cutting-edge research and development. We integrate the latest advancements in AI, machine learning, and educational technology to create an innovative and effective learning experience for students and educators
Area of specialization	Specializations for an AI-based classroom learning companion include expertise in AI/ML, data analysis, educational technology, and user experience design.
FYP ID	FYP-BCSM-F23-036

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*The candidates confirm that the work submitted is their own and appropriate credit has been given where reference has been made to the work of others

Plagiarism Free Certificate

This is to certify that, I Muhammad Rafeh S/D of Mahmood ul Hassan, group leader of FYP under registration no. FYP-BCSM-F23-036 at the Computer Science Department, The Superior University, Lahore. I declare that my FYP report has been checked by my supervisor.

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AI-Based Classroom Learning Companion

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APPROVAL

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Comments: _____

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Date:_____ Signature:_____

HEAD OF THE DEPARTMENT

Comments: _____

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Dedication

Our project, the AI-based Classroom Learning Companion, is dedicated to the pursuit of knowledge and the empowerment of learners. To the dedicated team behind this endeavor, your passion for education and commitment to harnessing the power of AI has shaped a tool that not only imparts information but inspires curiosity. May this dedication reflect the collective vision to redefine the educational landscape, fostering a future where learning knows no limits. Dedicated to the pursuit of smarter, more engaging, and inclusive education for all.

Acknowledgment

We extend our heartfelt gratitude to all contributors to the AI-Based Classroom Learning Companion. Special thanks to our dedicated team members for their unwavering commitment and expertise.

We appreciate the invaluable input from educators, shaping the educational framework. Thanks to Superior University for providing essential resources and a supportive environment.

Our users' feedback has been pivotal in refining the learning companion. This project's success is a collective effort, and we acknowledge every individual involved.

We express gratitude to our families, friends, and mentors for their continuous support. This project is a testament to collaborative innovation and dedication.

Thank you to everyone who played a role in realizing this trans-formative educational technology. Your belief in the project's potential has been a driving force.

Together, we've advanced the landscape of learning technology. Thank you for being an integral part of this journey.

Executive Summary

The AI-Based Classroom Learning Companion is an advanced educational tool utilizing artificial intelligence to transform traditional learning. By analyzing course materials, generating tailored test questions, and assessing student responses, it provides personalized insights into individual strengths and weaknesses, as well as common class knowledge gaps. Beyond assessments, the system identifies student interests, suggesting potential areas of excellence. Integrating open-source models for question preparation and career path recommendations, this software adapts to individual and collective learning needs. With its data-driven approach and integration of supplementary materials, the AI-Based Classroom Learning Companion aims to revolutionize education, creating a personalized and effective learning environment for students and educators.

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Chapter 1

Introduction

Chapter 1: Introduction

Our project is about making a cool computer program that schools, colleges, and universities can use. In this program, students can sign up for classes, watch videos to learn, take quizzes made by AI, and check their grades. Teachers can make these AI quizzes, see students' grades, give marks, and start new classes. If a class already exists, teachers can still teach it. This program helps everyone in school to learn and teach better.

1.1. Background

In the past, homework could only be completed in the classroom by teachers and pupils. However, learning may now take place at any time and anywhere thanks to computers and the internet. We are developing our AI Classroom Project for this reason. Whether learning is done at home or at school, we wanted to create a program that makes it enjoyable and simple. With the use of intelligent computer tools such as Laravel, OpenAI, React Native, and React, this program can assist teachers better organize their classes and help students learn in new ways. With the help of smartphones and computers, students can now view classes, complete quizzes, and view their grades, while teachers can monitor student progress and keep learning engaging for all.

1.2. Motivations and Challenges

Inspiration: Our goal in launching the AI Classroom Project was to increase accessibility to and engagement with education for everybody. With our software, students may study from any location; they are not limited to a classroom. Teachers can still impart their knowledge to additional students even when they are not in the same place. Our goal is to use technology to make education better for every kid.

Difficulties: Developing an application that works well on PCs and phones may be challenging. We must ensure that the videos, tests, and other materials in our program are simple for everyone to use and understand. It sometimes takes a lot of testing and experimenting to make sure everything works perfectly because of the tools we use, including Rails and React Native. We also need to make sure that teachers and students may easily use our application on any kind of computer or phone they own.

1.3. Goals and Objectives

Goals and Objectives for an AI-Based Educational Tool

Objectives: Regardless of background or ability, we want learning to be enjoyable and simple for everyone. By providing them with all the resources they require on their phones or computers, we hope to support them in achieving their academic goals. Our mission is to make teaching and monitoring student progress easier for educators.

Goal:

2. **Make Learning Accessible:** Whether students are at home or on the go, we want to ensure that they can learn at any time and from any location.
3. **Enhance Instructional Strategies:** Our goal is to assist educators by providing them with an AI-powered means of generating engaging tests and lessons.

1.4. Literature Review/Existing Solutions

Learning aids have previously been developed by countless programmers. We looked at a handful of these to get ideas for our project.

1. Online learning platforms: On websites, students can view videos and finish tests. With these, students may choose what they want to learn and progress at their own pace.

2. Classroom Management Software: Teachers can schedule classes and maintain grades with the help of a few tools. They make it easier for instructors to keep an eye on their sessions.

3. Educational Apps: There are apps that mimic games while teaching that can be downloaded for phones and tablets. Pupils find these entertaining and are inspired to learn more.

4. AI in Education: A couple more modern projects use AI to improve what students learn. AI is able to suggest lessons that a student should**.**

1.5. Gap Analysis

After looking over a number of learning support programs, we found a few that needed work. What we found out was this:

Not Always Easy to Use: Not all mobile devices, including tablets and phones, support all apps, making them difficult to use or malfunctioning.

Limited AI Use: While AI is used in some programs, its potential isn't fully realized. AI is capable of much more to enhance instruction and learning.

Not Enough Fun: Some educational tools have the potential to quickly bore students.

Lack of Flexibility: Many applications do not allow educators or learners to choose how best to use them. They have to follow the program's instructions.

The goal of our project is to solve these problems. Our application is made to be easily navigable across all platforms. AI is being utilized more and more to enhance student learning and teacher efficacy. In addition, we're enhancing our platform with fun and adaptability so that educators may instruct in a way that best fits their needs and students like using it.

1.6. Proposed Solution

We are creating a new program to increase student learning and instructor effectiveness. To be more precise, our program will

1. Works Anywhere: Our software is appropriate for use on PCs, tablets, and mobile phones, among other devices. It's easy to use whether you're at home or on the go.

2. Smart Learning with AI: Using AI, a form of advanced computer brain, our program generates lessons and quizzes that are uniquely customized for every learner. It guarantees that students understand everything and lets them learn at their own pace.

3. Interactive and Fun: To make our program more enjoyable for you to study in, we've integrated illustrations and charts. It won't be boring because we want you to enjoy learning.

4. Easy for Teachers: For teachers, the chores of grading, making quizzes, and keeping track of student achievement are all easy. They can also use the program to make teaching simpler and more enjoyable.

5. Adaptable for Everyone: Teachers and students are free to choose the subjects and teaching pedagogies they choose to employ with our software. Given its adaptability, it can satisfy the needs of all users.

Our new approach aims to raise student satisfaction in the classroom and enhance learning.

It will, in our opinion, assist educators in imparting knowledge how they see fit and motivate learners to achieve success.

1.7. Project Plan

We have some important steps to follow that will help us make our new learning program. Here's our plan:

- **The Program Design:** Initially, we will sketch pictures and make blueprints of how our programme should be structured and operate with regards to the needs and desires of students and teachers.

- **The Program Construction:** Then, we will employ special computer tools such as Rails, React Native, and AI from OpenAI in order to build our project. We are going to write codes making sure they are going well both on computers' and mobile phones' screens.

- **Test Everything:** After we build it, let's make sure everything is working 100%. We will request several students or teachers who can try to use it firstly in order to learn whether they like it or not as well as if everything works accordingly.

- **Troubleshooting:** If there are any issues found, we shall mend them so that the program could turn out user friendly while fun using.

- **How To Use It Will Be Taught:** Lastly, students and teachers shall be shown how this program operates so that they know how one logs in; watches videos; takes quizzes; enjoys other great things among others.

Let's Launch The Program: As soon as all is set up properly and smoothly running then our program will be put into all schools for every student worldwide together with their teachers to adopt use it respectively.

1.7.1. Work Breakdown Structure

Creating our learning program requires completing a number of tasks. We dissect them into the following smaller steps:

1. Organizing:

• Gather Ideas: Ask teachers and students for suggestions or anything else they might need.

• Create the Layout: Sketch out the appearance of the website and app.

2. Progress:

• Set Up Tools: Install software tools that you can use, such Rails and React.

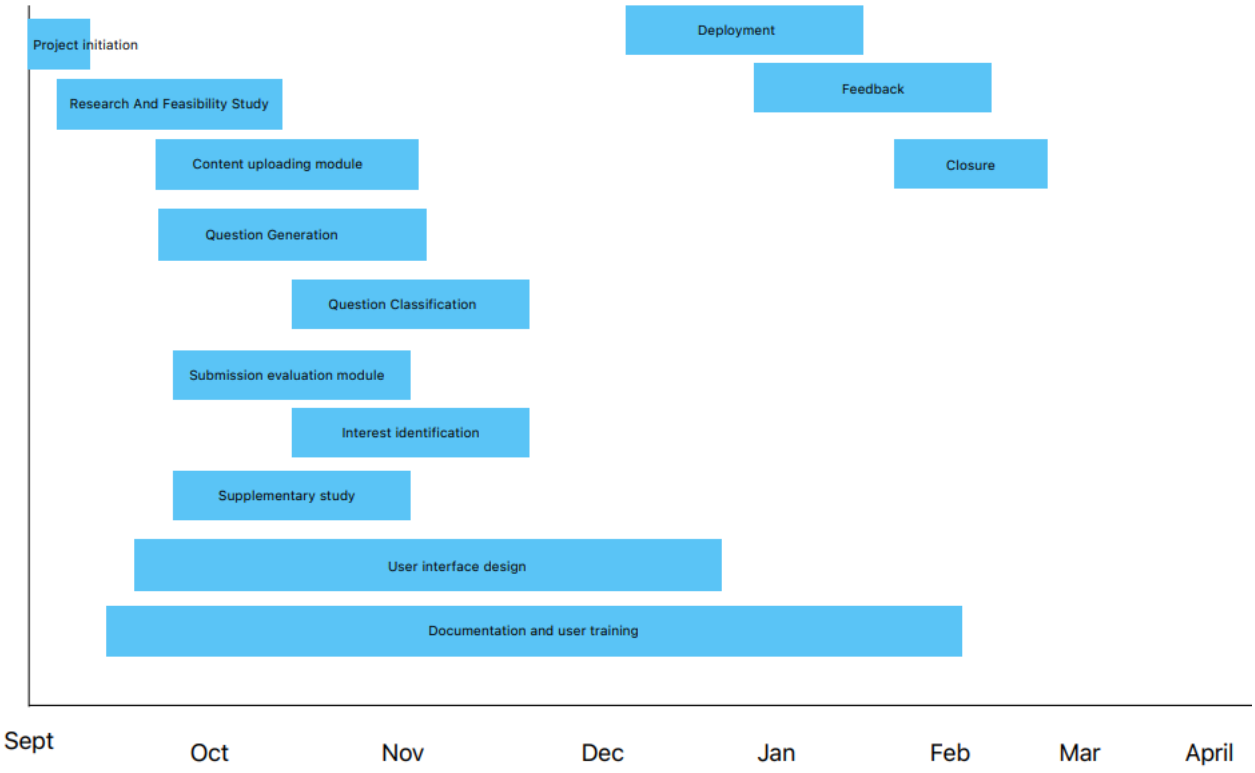
• Write Code: Take the quizzes and video parts, for instance.

• Develop Features: Lastly, you can incorporate functions like grade access, quizzing, and video viewing.

3. Evaluation:

• Developer testing: This is how our team tests to see if everything functions properly or not.

1.7.2. Gantt Chart



1.8. Empathy Map

Says: • "We need a more efficient way to assess student understanding." • "Identifying and addressing individual learning gaps is crucial." • "Customized study resources can greatly benefit our students."	**Thinks:** • "Effective assessment tools can enhance our teaching methods." • "AI technology can provide valuable insights into student performance." • "Personalized learning experiences can improve student outcomes."
Does: • Spends time curating and organizing course materials. • Analyzes student submissions and provides feedback. • Strives to create engaging and tailored learning experiences.	**Feels:** • Concerned about students' academic progress and success. • Optimistic about the potential of AI to revolutionize education. • Frustrated when faced with the challenge of individualized support.

Chapter 2

Software Requirement Specifications

Chapter 2: Software Requirement Specifications

2.1. Introduction

2.1.1. Purpose

The AIM of the AI-Based Learning Companion is to revolutionize the existing educational platform through artificial intelligence and sophisticated technology. It is designed to provide a unique and growing educational experience for each student through safe login, thematic quiz creation. For teachers, it streamlines the course management process by offering interest-based registration, in-depth analysis, and legally validated documentation.

2.1.2. Document Conventions

Guidelines on Formatting the Project Documentation for the AI-Based Learning Companion to ensure that all the communications with the developers are formal and clear, and that there is not information overload. Under this guideline, the headings are in font 15, and body text headers in font size 12. Time New Roman is the font for the entire document.

2.1.3. Intended Audience and Reading Suggestions

AI-Based Learning Companion – revolutionize education with the use of artificial intelligence and innovative technology. The AI-Based Learning Companion provides a customized, flexible learning experience for students that comes with protected login, an automatic quiz generator, and numerous communication modes. As it is a powerful tool for school administrations, the new program makes course administration easy with custom registration and extensive analytical data on students . Secure documentation storage is included.

2.1.4. Product Scope

The AI-Powered Learning Assistant seeks to develop a sophisticated educational environment that tailors the learning journey to individual students and educators. Key features include:

- Registration and secure login for users
- Recommending Solutions and open source education material

- Interactive quizzes and evaluations
- Detailed analytics and more

2.2. Overall Description

2.2.1. User Classes and Characteristics

Learners. Attributes. They include different age categories and prior experience with studies. Although some of them are new to learning, many of them have precognitions, preferences, and attitudes toward learning. Engagements. Registration for courses, watching educational videos, working on quizzes. Interactions with the chatbot . Writing reviews on each course and the studying process in general. * Educators. Attributes. They are experts in certain fields or have specific skills while prioritizing quality education. Engagements . Creation of tasks, monitoring of students, and evaluation of their performances. Building courses about certain skills and filling them with content. Assessment of quizzes and an evaluation of learners' achievements. Application of analytics. * Software developers. Attributes. They are information technology professionals who are focused on building and operating systems. Profound knowledge of different programming languages and artificial intelligence tools.

2.2.2. Operating Environment

Our learning program is designed to work in many different places where students and teachers use computers and mobile devices. Here are the main parts of our operating environment:

1. **Devices:** Our program works on computers, smartphones, and tablets. This means you can use it whether you have a Windows computer at school, a Mac at home, or an Android or iOS phone in your pocket.
2. **Internet:** You need to be connected to the internet to use our program. This is because it stores all the information online, so you can access it from anywhere.
3. **Software Tools:** We are building our program with some cool tools. We use Laravel to manage the data on the web, React Native to make the mobile app, and React for the web app. These tools help make sure that everything you do in the program is smooth and fast.

4. **Users:** Our program is made for students and teachers in schools, colleges, and universities. It's easy enough for everyone to use, even if you're not very good with computers.
5. **Safety and Security:** We make sure that our program is safe to use. Only you can see your information, like your grades or the quizzes you take. Teachers can see how their students are doing, but all the data is protected so no one else can see it.

Our goal is to create a safe, easy, and fun environment where you can learn a lot and teachers can help you better.

2.2.2.1. Functional Requirements

Our learning program needs to do several important things to help students and teachers. Here's what it needs to do:

1. **Student Access:**
    - **Log In:** Students can log into their own account with a username and password.
    - **View Classes:** Students can see a list of all the classes they are enrolled in.
    - **Watch Videos:** Students can watch educational videos related to their classes.
    - **Take Quizzes:** Students can take quizzes that the program creates using AI to help test what they've learned.
    - **See Grades:** After taking quizzes, students can look at their grades to see how well they did.
2. **Teacher Access:**
    - **Log In:** Teachers can log into their account to access their tools.
    - **Create Quizzes:** Teachers can make new quizzes for their students using the AI to make them challenging and fun.
    - **View and Assign Grades:** Teachers can look at how students are doing and put in grades for their work.
    - **Create and Manage Classes:** Teachers can start new classes and manage the ones they already have.
3. **AI Features:**
    - **Generate Quizzes:** The AI should be able to create quizzes based on the subjects teachers are teaching.

- **Suggest Learning Materials:** The AI can suggest videos and other learning materials that are good for each student.

4. **Compatibility:**
    - **Work on All Devices:** Our program must work well on computers, tablets, and smartphones.
    - **Be Easy to Use:** The design should be simple so that everyone, even those who aren't good with technology, can use it easily.
5. **Security:**
    - **Protect Information:** The program must keep all student and teacher information safe and private.

These are the main things our program needs to do so that everyone can learn and teach well.

2.2.2.2. Non-Functional Requirements

Our learning program also needs to meet some special requirements that make sure it works really well and is nice to use. Here's what we need:

1. **Speed:**
    - **Fast Loading:** The program should load quickly on all devices, so you don't have to wait a long time to start learning or see your grades.
2. **Reliability:**
    - **Always Available:** Our program should work all the time, so you can learn whenever you want without any problems.
3. **Usability:**
    - **Easy to Navigate:** The program should be easy for everyone to understand and use. You should be able to find what you need without help.
4. **Scalability:**
    - **Handle Lots of Users:** Our program should work well even if lots of students and teachers are using it at the same time.
5. **Security:**
    - **Safe and Secure:** Our program must protect all your personal information and make sure only you and your teachers can see your grades and work.
6. **Compatibility:**

- **Works with Different Devices:** The program should work well whether you are using a computer, a tablet, or a smartphone.

7. **Maintainability:**
    - **Easy to Update:** We should be able to fix any bugs or add new features to the program without causing problems for users.

These are the special rules our program follows to make sure it's always working right and is easy and safe for you to use.

2.3. Domain Requirements

Our learning program has some specific needs based on where it's being used and who is using it. Here are the important rules and needs for our program:

1. **Educational Standards:**
    - **Follow School Rules:** Our program must match what schools, colleges, and universities expect from learning tools. It should help students meet their learning goals.
    - **Cover All Subjects:** The program should be useful for many different subjects, not just one or two.
2. **Technology Needs:**
    - **Use Modern Tools:** We are using up-to-date technology like Laravel for the web, React Native for mobile apps, and AI to make the program smart and helpful.
    - **Internet Access:** Since our program works online, users need to have good internet access to use all its features.
3. **User Requirements:**
    - **For Students:** Students should find the program engaging and helpful for learning on their own or with teacher guidance.
    - **For Teachers:** Teachers need tools within the program to easily create content, track student progress, and manage classes.
4. **Accessibility:**
    - **Easy for Everyone:** The program must be easy for all students to use, including those who might need extra help like students with disabilities.
5. **Legal and Compliance:**

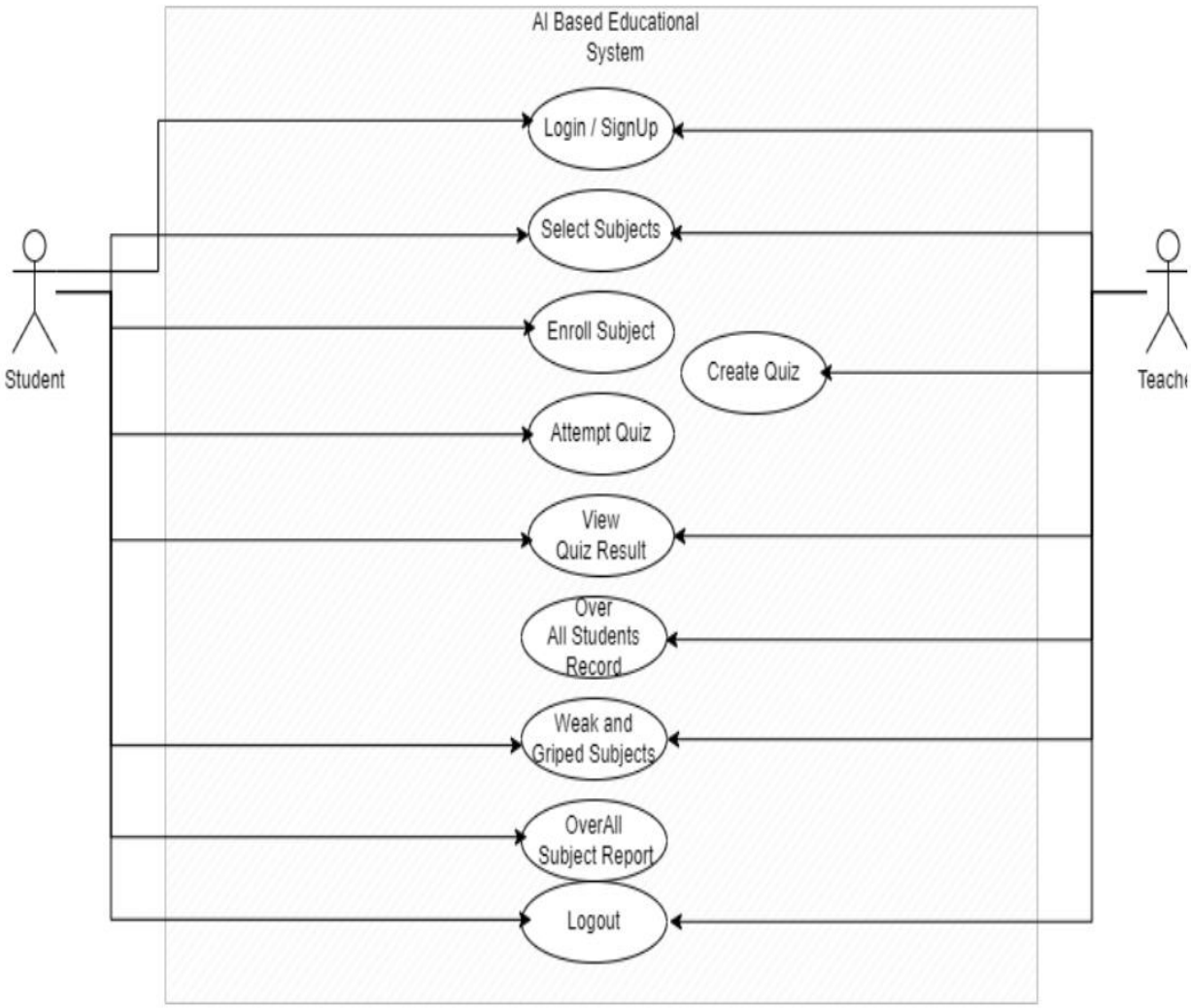
- **Privacy Laws:** We must follow laws that protect student information and ensure their privacy is respected.

These domain requirements make sure our program does what it's supposed to do for everyone who uses it, in a way that's safe, legal, and really helpful.

Chapter 3

Use Case Analysis

3.1. Use Case Model



3.2. Use Cases Description

Our learning program is designed to help students and teachers in many different ways. Here are some of the main ways people will use our program:

1. **Student Use Cases:**
    - **Logging In:** Students will start by logging into their account with a username and password.
    - **Viewing Classes:** Once logged in, students can see a list of all the classes they are taking.
    - **Watching Tutorial Videos:** Students can watch videos that teach them about different topics in their classes.

- **Taking Quizzes:** Students can take quizzes that our smart AI makes, to see how much they have learned.
- **Checking Grades:** After quizzes, students can look at their grades to see how well they did.

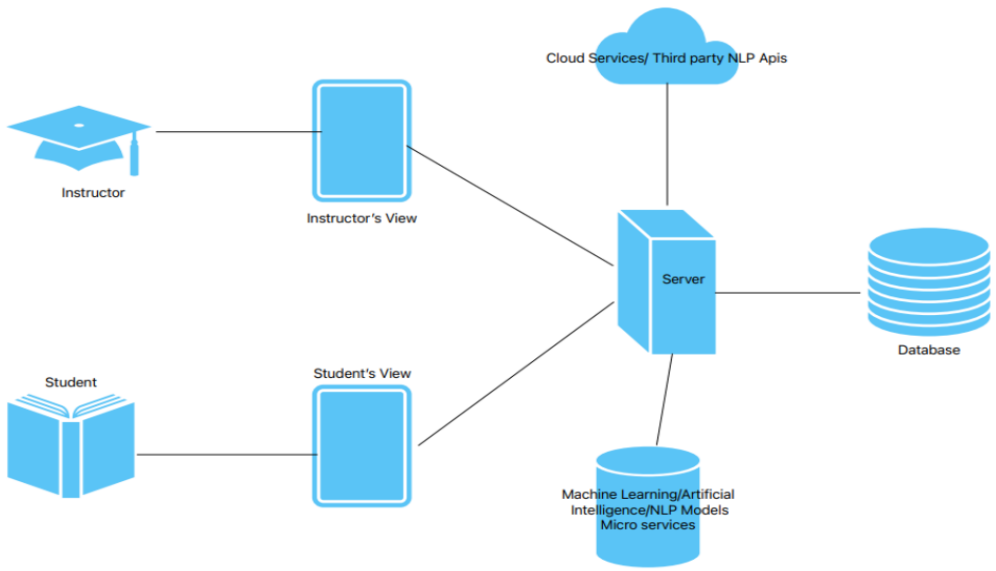
2. **Teacher Use Cases:**
    - **Logging In:** Teachers also log in with their account details to access their tools.
    - **Creating Quizzes:** Teachers can use the program to make new quizzes for their students with the help of AI.
    - **Viewing Student Progress:** Teachers can see how each student is doing and check their grades.
    - **Assigning Grades:** Teachers can give grades based on the quizzes and other assignments.
    - **Creating and Managing Classes:** Teachers can start new classes and add or change the information for the classes they already teach.
3. **Administrative Use Cases:**
    - **Adding New Courses:** School administrators can use the program to add new courses to the school's curriculum.
    - **Monitoring Usage:** Administrators can check how much students and teachers are using the program and make sure it's helping in the best way.

Each of these use cases helps make sure that our program is doing a great job for students learning and teachers teaching. It makes schoolwork easier and more fun!

Chapter 4

System Design

4.1. Architecture Diagram



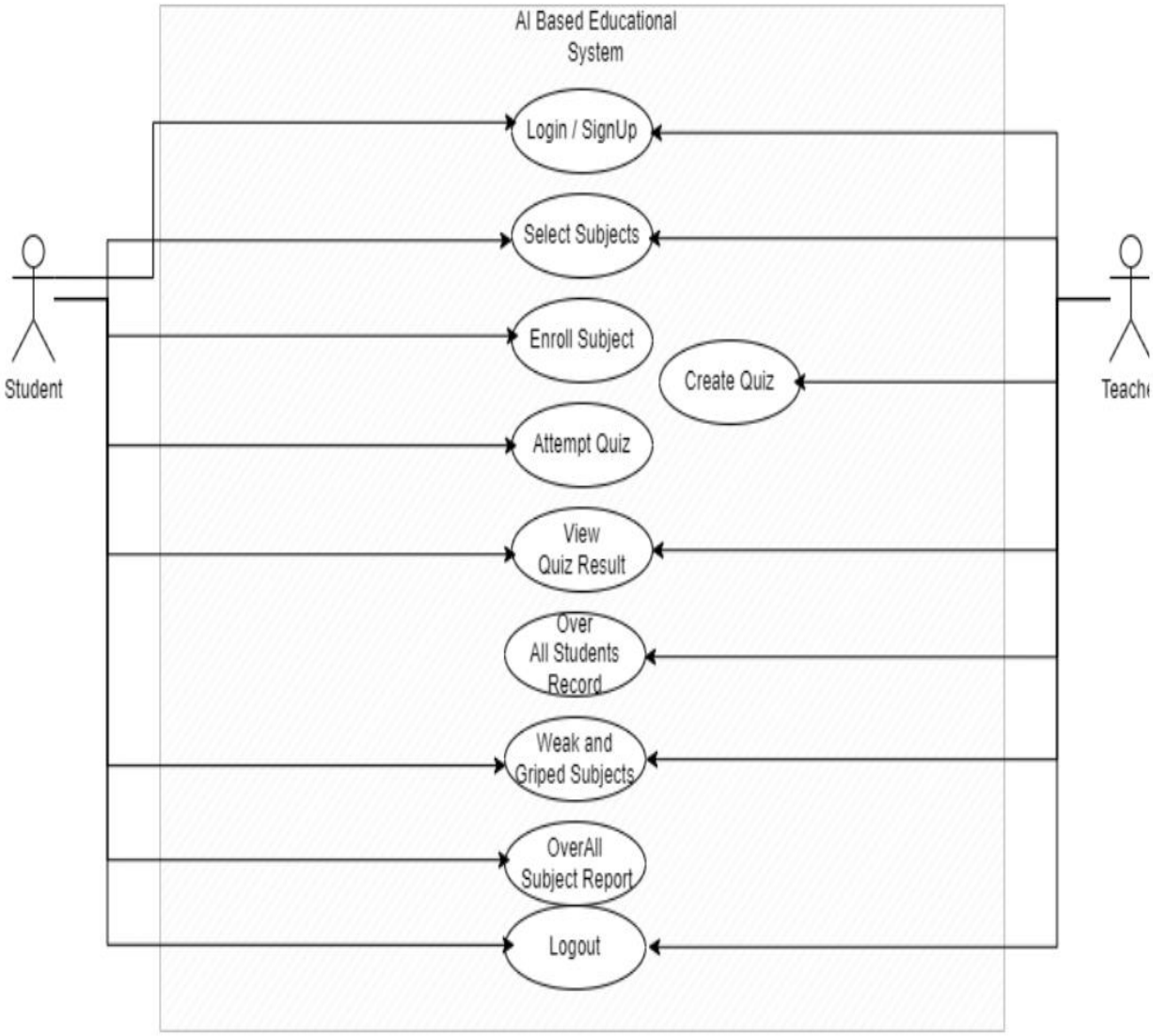
4.2. Domain Model

Our learning program is like a big toolbox that helps students and teachers do their schoolwork online. To understand how everything works together, let's talk about the main parts of our program:

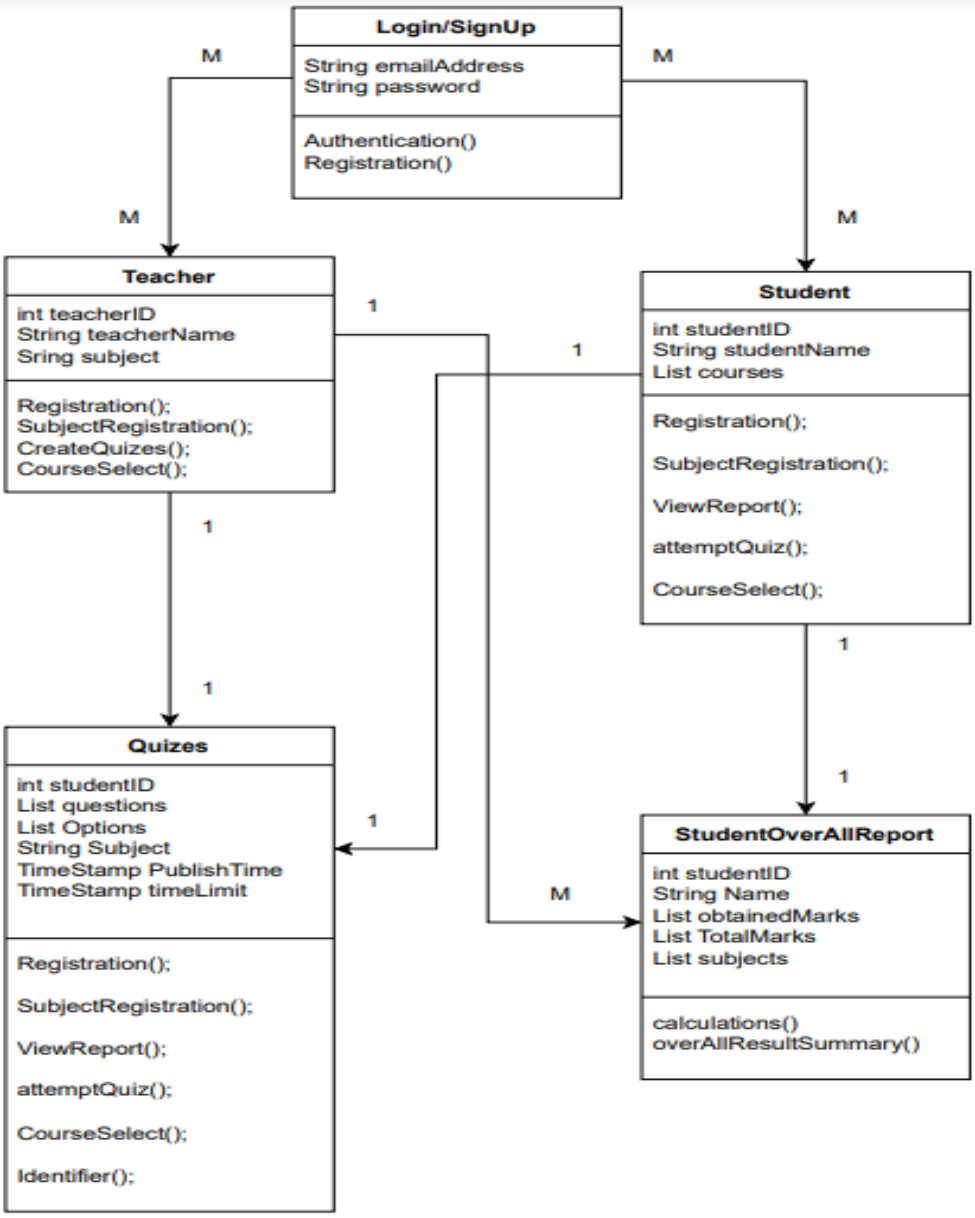
1. **Users:**
    - **Students:** They log in to learn, watch videos, take quizzes, and check their grades.
    - **Teachers:** They log in to make quizzes, see how students are doing, and manage classes.
    - **Administrators:** They make sure everything in the program works right and can add new courses or features.
2. **Courses:**
    - These are the different subjects that students can learn about in our program. Each course has its own videos, quizzes, and materials.
3. **Quizzes:**
    - These are tests that the AI creates to help students learn better. Quizzes help teachers see how much students have learned.
4. **Grades:**
    - These are scores that students get after they take quizzes or complete assignments. Grades help students know how well they are doing.
5. **Videos and Materials:**
    - These are learning tools within each course. Videos help students understand topics better, and materials might include things like worksheets or readings.
6. **AI (Artificial Intelligence):**
    - This is the smart part of our program that helps make quizzes and suggests videos that are good for each student's learning.

Our domain model shows how all these parts connect and work together. It's like a map that helps us see how each piece helps students and teachers in our program.

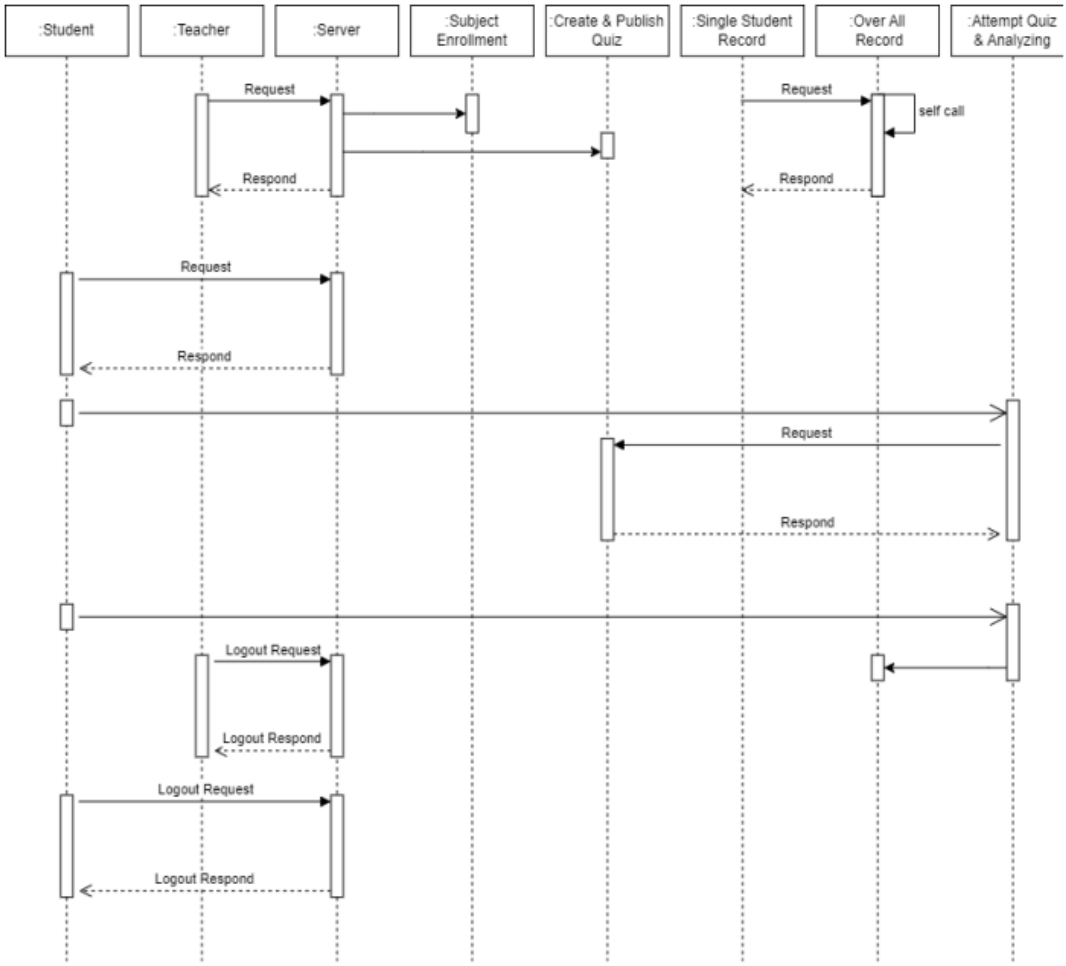
4.3. Entity Relationship Diagram with Data Dictionary



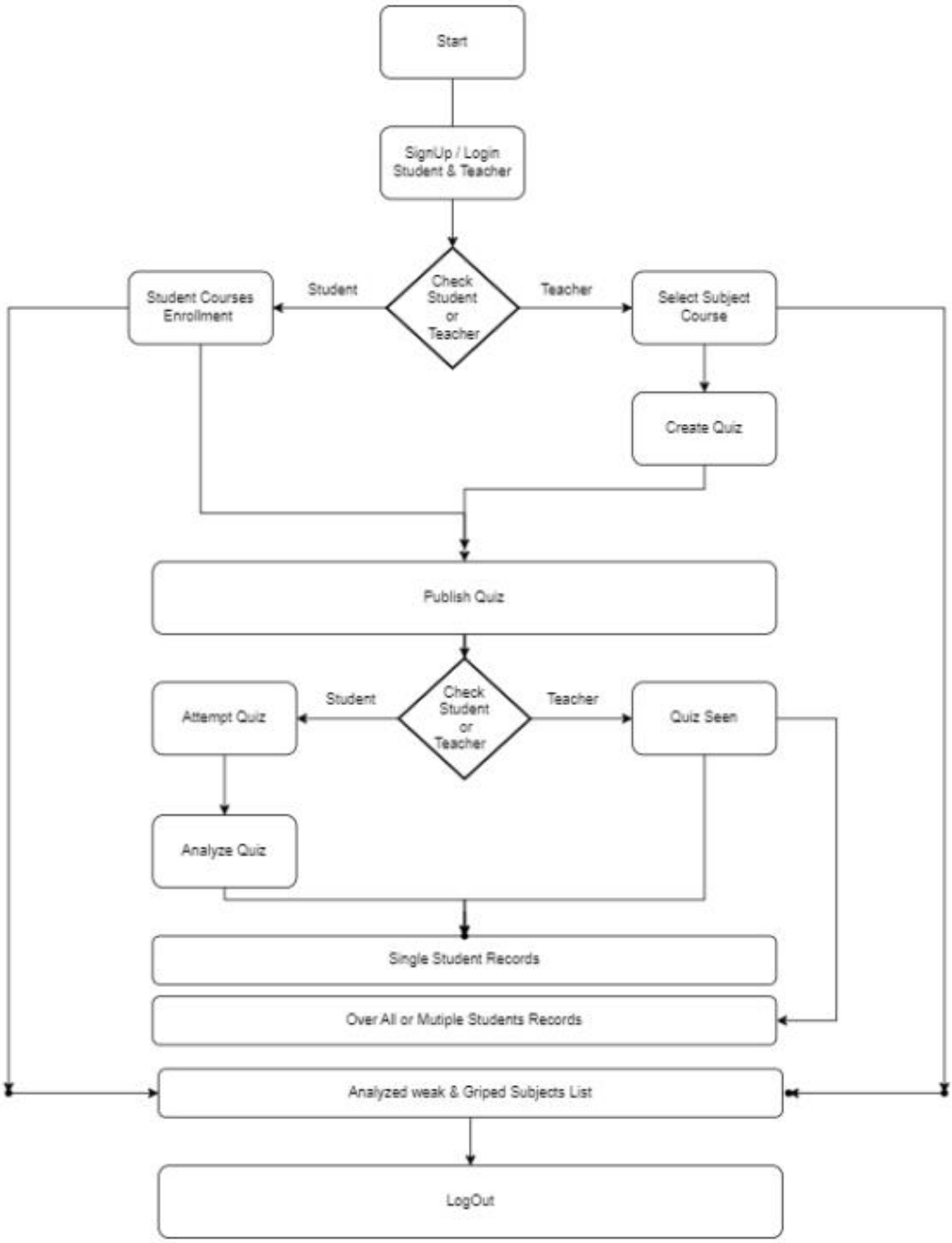
4.4. Class Diagram



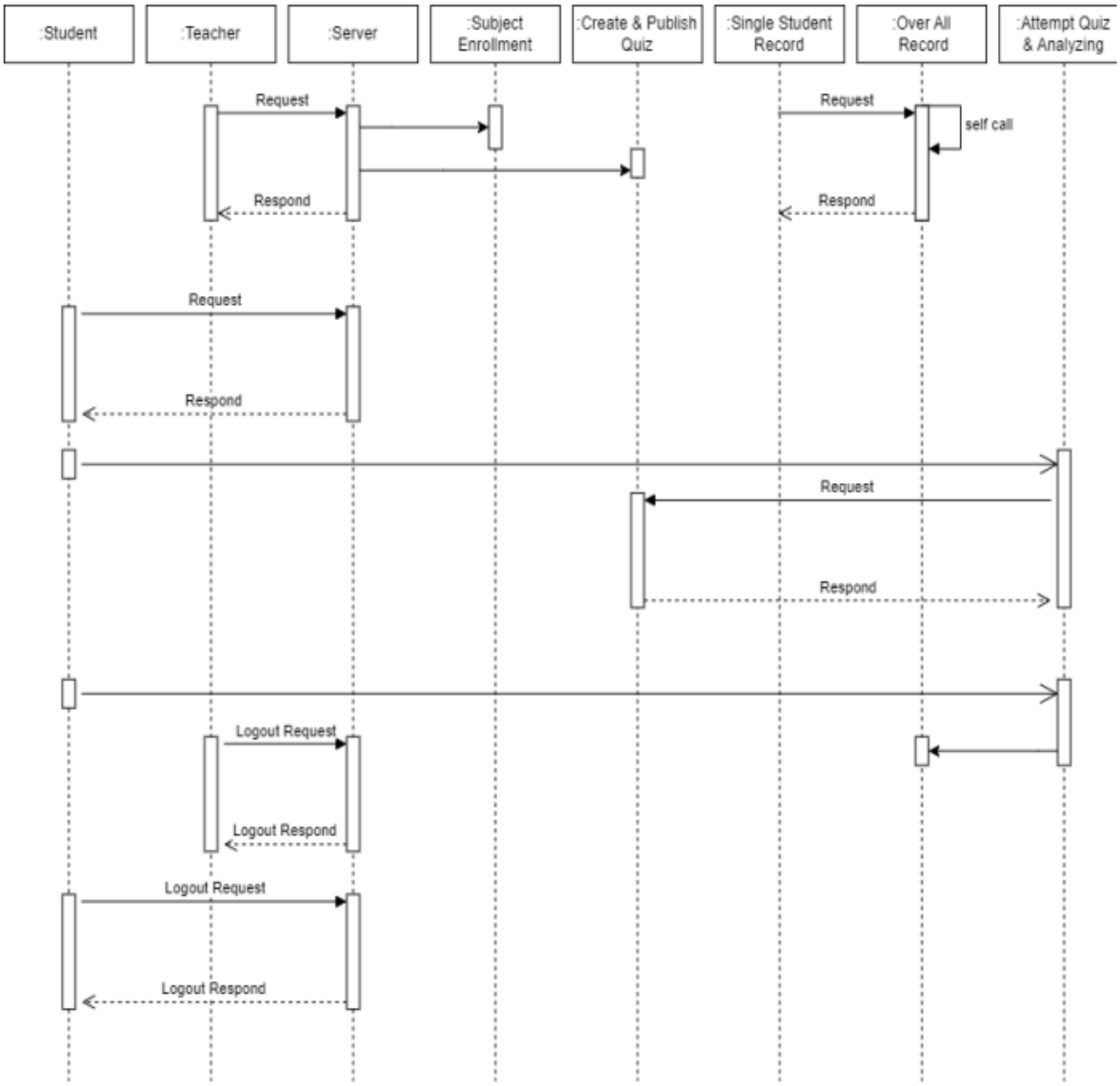
4.5. Sequence / Collaboration Diagram



4.6. Activity Diagram



Data Flow diagram



Chapter 5

Implementation

Chapter 5: Implementation

Implementing our AI classroom project means actually building and setting up our program so students and teachers can start using it. Here's how we do it:

1. **Writing the Code:**
    - First, we use special computer languages to write instructions that tell the program what to do. We use tools like Laravel for web pages, React Native for mobile apps, and React for other web parts.
2. **Setting Up the Database:**
    - We need a place to store all the information like quiz results, student grades, and video lessons. So, we set up a database that keeps all this data safe and organized.
3. **Creating User Interfaces:**
    - This is about making the screens that students and teachers see when they use the program. We design these screens to be simple and fun to use, with buttons and menus in places that make sense.
4. **Integrating AI Components:**
    - We add the AI parts that help make quizzes and suggest learning materials. This AI helps customize the learning experience for each student.
5. **Testing Everything:**
    - Before anyone starts using the program, we test it a lot. We make sure there are no mistakes and that everything works as it should. Sometimes we ask a few students and teachers to try it and tell us if they find any problems.
6. **Launching the Program:**
    - Once everything works well and everyone is happy with it, we make the program available to all schools. We might have a special day where we show everyone how to log in and start learning.
7. **Getting Feedback and Making Changes:**
    - After the program is out, we keep listening to what students and teachers say about it. If they have problems or want new features, we work on adding those to make the program even better.

Implementing our program takes a lot of careful work, but it's exciting to see it come to life and help students learn in new and fun ways!

5.1. Important Flow Control/Pseudo codes

The codebase incorporates both functional components and class components. Control over the primary operations of the application is managed by the App component. This component is responsible for initializing the Redux store and handling the rendering of the Routes component. In turn, the Routes component employs React Router to establish the various routes within the application, ensuring that the correct component is displayed in response to the active URL.

5.2. Components, Libraries, Web Services and stubs

Flow control in our program is like giving directions on how to get from one place to another. Pseudo code helps us plan these directions by writing them down in simple steps. Here's how some of the main parts of our program work:

1. **Logging In:**
    - **Check Username and Password:**
        - If the username and password are correct, go to the user's homepage.
        - If they are wrong, show a message to try again.

```
IF username and password are correct
   GO to homepage
ELSE
   SHOW "try again" message
```

2. **Taking a Quiz:**
    - **Start Quiz:**
        - Show the first question.
        - When a student answers a question, show the next one.
        - If there are no more questions, show the quiz results.

```
START quiz
FOR each question in the quiz
   SHOW question
   GET answer
   IF there are more questions
      CONTINUE to next question
```

```
ELSE
  SHOW results
```

3. **Viewing Grades:**
   - **Select Course:**
     - Show all the courses the student is taking.
     - Let the student choose a course to see grades.

```
SHOW all courses
WHEN student selects a course
  DISPLAY grades for that course
```

4. **Creating a Quiz (Teacher):**
   - **Add Questions:**
     - Let the teacher write a question.
     - Add possible answers.
     - Repeat until the teacher finishes making the quiz.

```
START creating quiz
WHILE teacher wants to add questions
  WRITE question
  ADD answers
  ASK "Add another question?"
  IF no
    END creating quiz
```

These steps help make sure that our program knows what to do next based on what you tell it. It's like following a recipe to bake a cake—each step needs to be clear so the cake turns out just right!

5.3. Deployment Environment

The deployment environment is where our program lives and works so that students and teachers can use it. Here's how we set up our program to make sure it runs smoothly:

1. **Web Servers:**
   - We use special computers called web servers that store our program and send it to your computer or phone when you want to use it. These servers are always on and connected to the Internet.
2. **Mobile and Web Apps:**

- Our program works on both the web and as an app on your phone or tablet. This means you can use it in your web browser like Chrome or Safari, or you can download an app from the App Store or Google Play.

3. **Cloud Services:**
   - We use cloud services, which are also special computers on the Internet, to keep all the data safe. This includes your quiz results, the videos you watch, and your grades. The cloud makes sure you can access your data from anywhere and that it's always backed up, so it won't get lost.
4. **Updates and Maintenance:**
   - Sometimes we need to update the program to add new features or fix any problems. When we do this, we first test everything in a special testing environment to make sure it all works right. Then we update the real program that everyone is using.
5. **Security Measures:**
   - We put strong security in place to protect your information. This includes things like passwords and encryption, which is a way of scrambling data so only the right people can read it.

Our deployment environment is set up to be strong and reliable so that you can always use the program without any trouble, and so that your information stays safe.

5.4. Tools and Techniques

To build our learning program, we use some cool tools and special techniques to make sure everything works well and is fun to use. Here's what we use:

1. **Programming Languages:**
   - **Laravel:** This is a tool that helps us build parts of our website. It's like using building blocks to make sure all the parts of our site work together.
   - **React and React Native:** These tools help us make our program look nice and work fast on both computers and mobile phones. React is used for websites, and React Native is for apps on phones and tablets.
2. **AI with OpenAI:**
   - We use AI, or artificial intelligence, from a company called OpenAI to help make quizzes and suggest videos that are just right for each student's learning. It's like having a smart helper in our program.

3. **Databases:**
    - **MySQL:** This is where we keep all the information, like quiz scores and video links. Think of it as a big filing cabinet where everything is neatly organized so we can find it quickly when we need it.
4. **Development Tools:**
    - **GitHub:** We use GitHub to keep all our program's code safe and to share it with other team members. It's like a library of all the work we've done, where we can also fix mistakes and improve things.
    - **Visual Studio Code:** This is a tool where we write and check our code. It's like a smart notebook that helps us write better and find any mistakes.
5. **Testing Tools:**
    - **Jest:** This tool helps us test our code to make sure it works right before anyone uses it. It's like practicing a play before performing it in front of an audience.
    - **Selenium:** We use Selenium to pretend we are users to see how well our program works. It clicks on buttons and fills out forms just like a student or teacher would, to make sure everything is easy to use and working as expected.
6. **Security Measures:**
    - **SSL Certificates:** These are like ID badges for our program that prove it is safe and secure for you to use. They help protect your information from anyone who shouldn't see it.

By using these tools and techniques, we make sure our program is safe, fast, and a fun place to learn!

5.5. Best Practices / Coding Standards

When we build our learning program, we follow some important rules to make sure everything works well and is easy to use. Here are some of the best practices and coding standards we use:

1. **Keep It Simple:**
    - We try to make our code simple and clear. This means writing our program in a way that is easy to understand, just like writing a clear and easy-to-read

story. This helps us and anyone else who looks at our code to know what it does and how to fix or change it if needed.

2. **Use Comments:**
    - We write comments in our code, which are like little notes that explain what different parts of our program do. These notes help others (and us later on) understand why we wrote the code the way we did. It's like leaving a sticky note in a book to remind you why something is important.
3. **Test Often:**
    - We check our work by testing our code a lot to find any mistakes or problems. It's like proofreading your homework before turning it in. This helps us make sure that everything in our program works perfectly when students and teachers use it.
4. **Stay Consistent:**
    - We use the same style of writing code throughout our project. This includes how we name things and how we organize our code. Consistency makes our program easier to read and maintain, kind of like following the same rules in a game.
5. **Secure Everything:**
    - We make sure our program is safe for students and teachers to use. We protect it from people who might try to break in and cause problems. It's like making sure the doors and windows in your house are locked.
6. **Reuse Code:**
    - When parts of our program can be used more than once, we write them in a way that lets us reuse them. This saves time and effort, just like using a template for a school project.
7. **Keep Updating:**
    - We keep our tools and techniques up-to-date. This means we use the latest and best tools to build our program, ensuring it stays modern and works well.

By following these best practices and coding standards, we make sure our learning program is strong, secure, and fun for everyone to use!

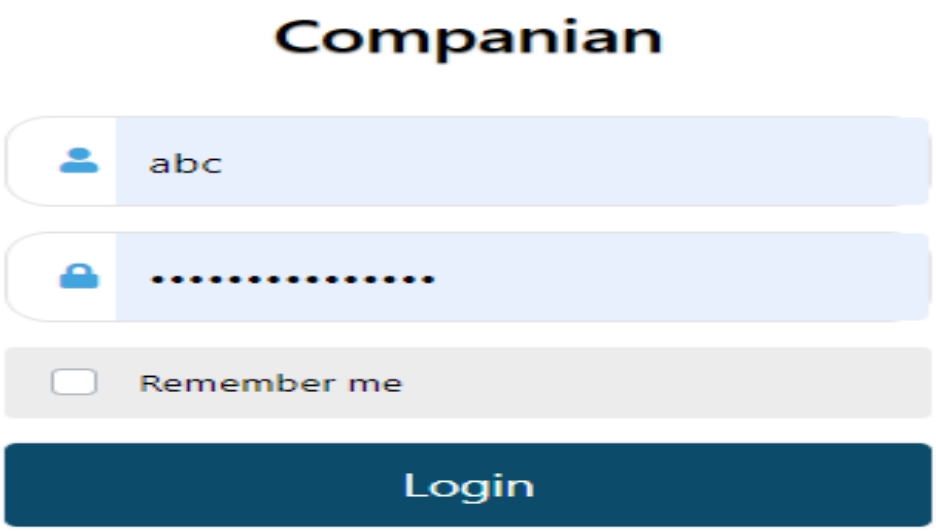
5.6. Version Control

Version control is like keeping a magic diary for our program where every change we make is recorded. It helps us remember what we did and when we did it, so if something goes wrong, we can go back and fix it. Here's how we use version control:

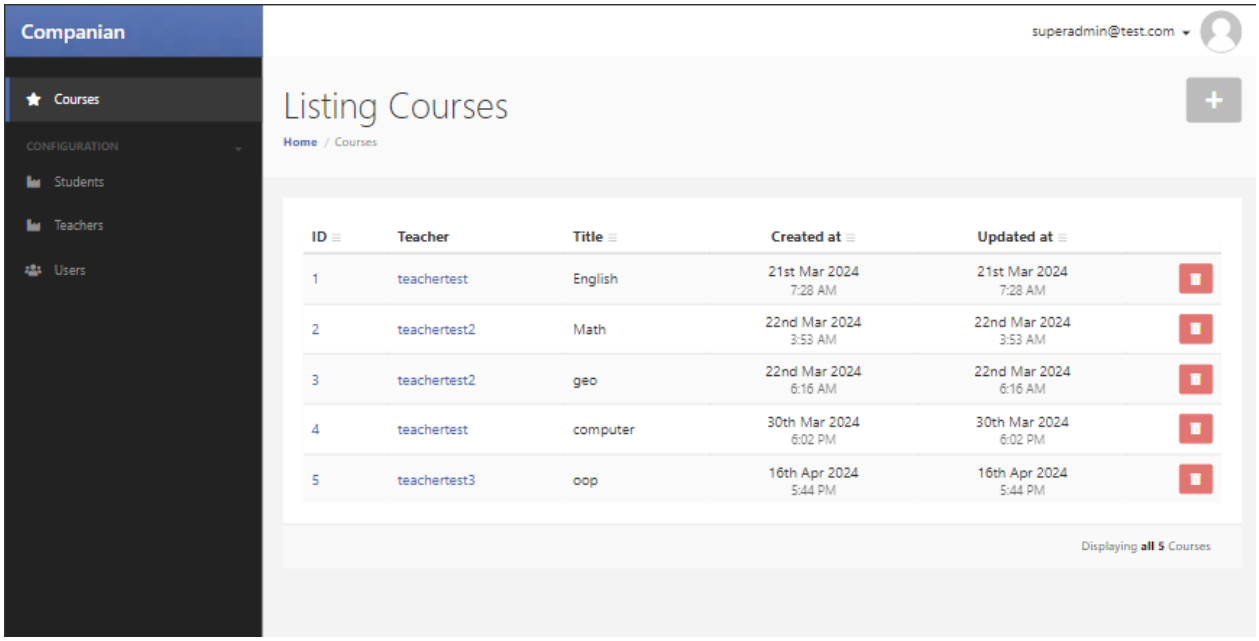
1. **Save Changes:**
   - Every time we make a change to our program, like adding new features or fixing a mistake, we save it. It's like writing down what you did in your school project each day.
2. **Work Together:**
   - Our team has many people working on the program at the same time. Version control helps us make sure that we don't mess up each other's work. It's like playing a team sport where everyone knows their position and plays together without getting in each other's way.
3. **Keep Old Versions:**
   - We keep all the old versions of our program, so if the new version has a problem, we can look at the old one to see what changed. It’s like having a backup of your favorite toy—you know you’ll have something to play with even if the new one breaks.
4. **Track Who Did What:**
   - Version control lets us see who made which changes. This is helpful if we need to ask questions or understand why something was done a certain way. It's like signing your artwork, so everyone knows you made it.
5. **Tools We Use:**
   - We use a tool called **GitHub** to manage our version control. It keeps all our code in one place on the internet, and anyone on our team can see it and make changes. It's like a shared folder for your class where everyone can put their work and look at each other's.

By using version control, we make sure our program is always getting better and that we can always fix any mistakes easily. It keeps everything organized and safe!

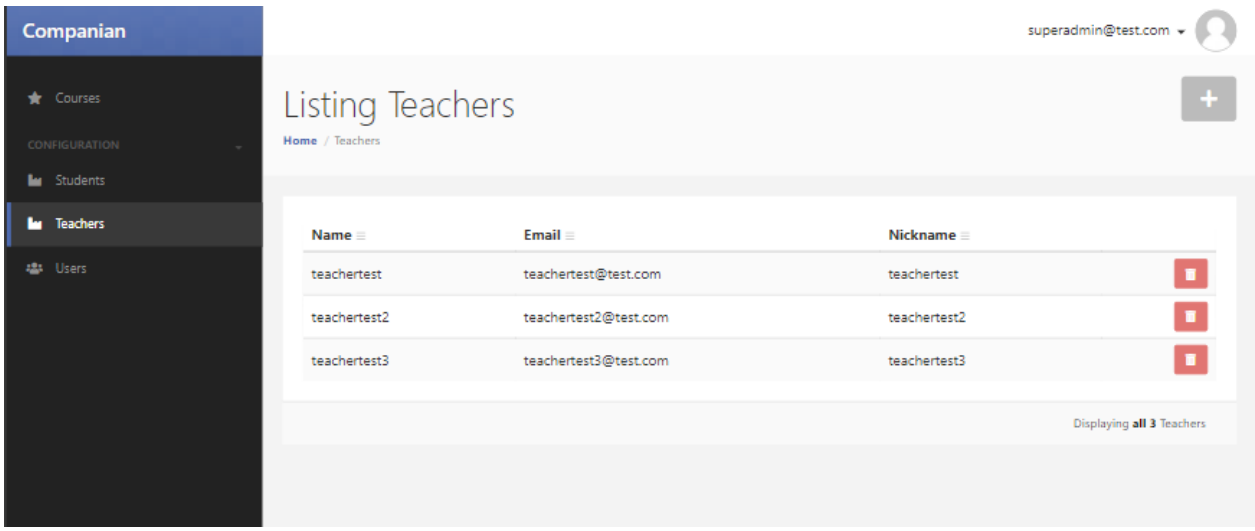
5.7. Login Page



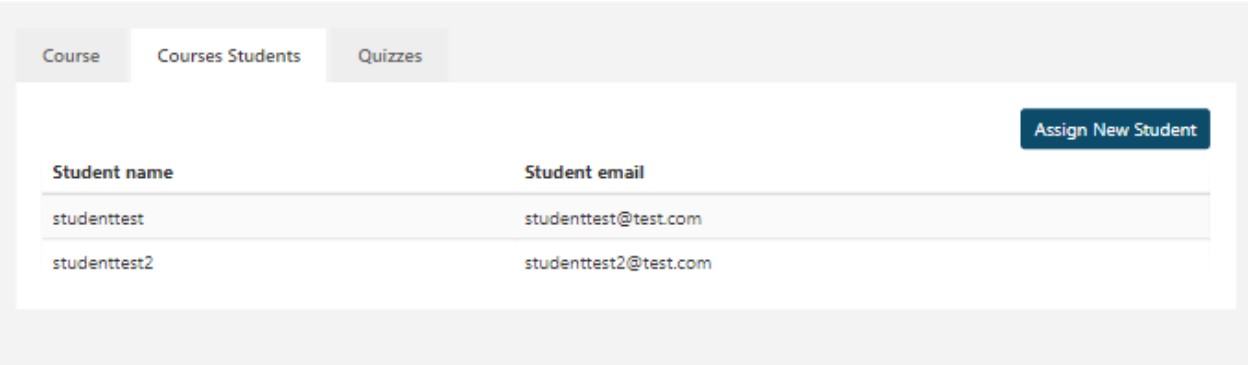
5.7.1. Register Courses



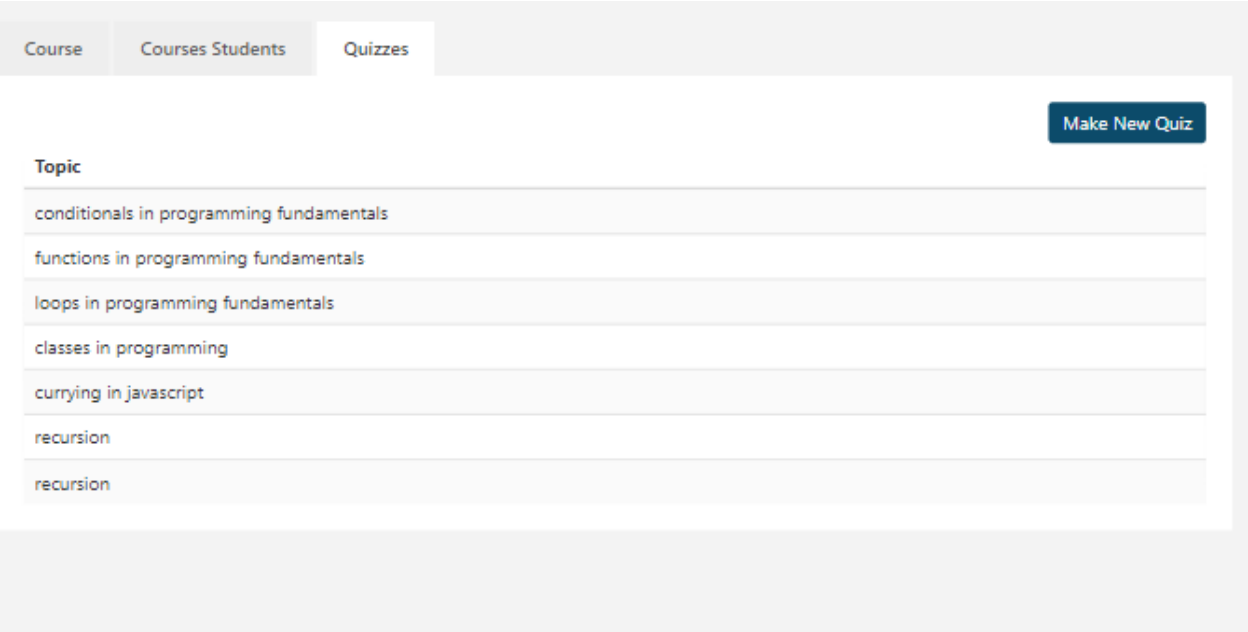
5.7.2. List of Teacher



5.7.3. Student Courses Enroll



5.7.4. Quizzes History



5.7.5. Quiz Generate

Quiz | Submissions | Submissions Chart | Quiz Insights

Topic

functions in programming fundamentals

Questions

1:What is a function in programming?

1 : A variable

2 : A reusable block of code

3 : A loop

4 : A data type

2:What is the purpose of a function?

1 : To store data

2 : To perform a specific task

3 : To display output

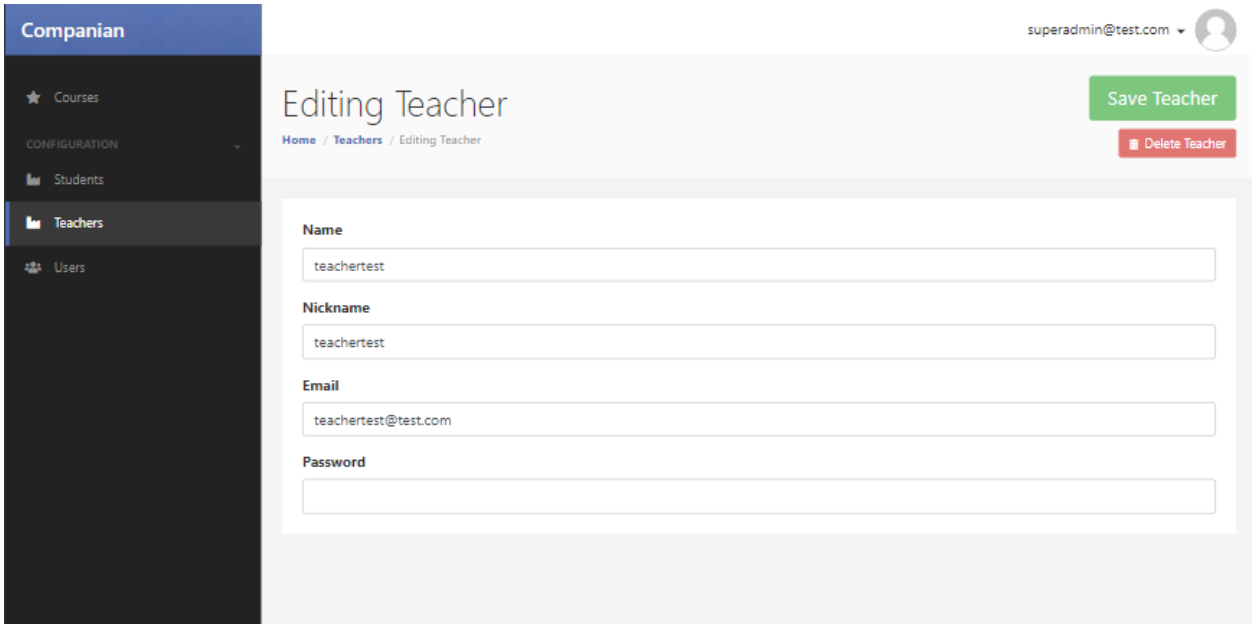
4 : To create variables

3:Which keyword is used to define a function in JavaScript?

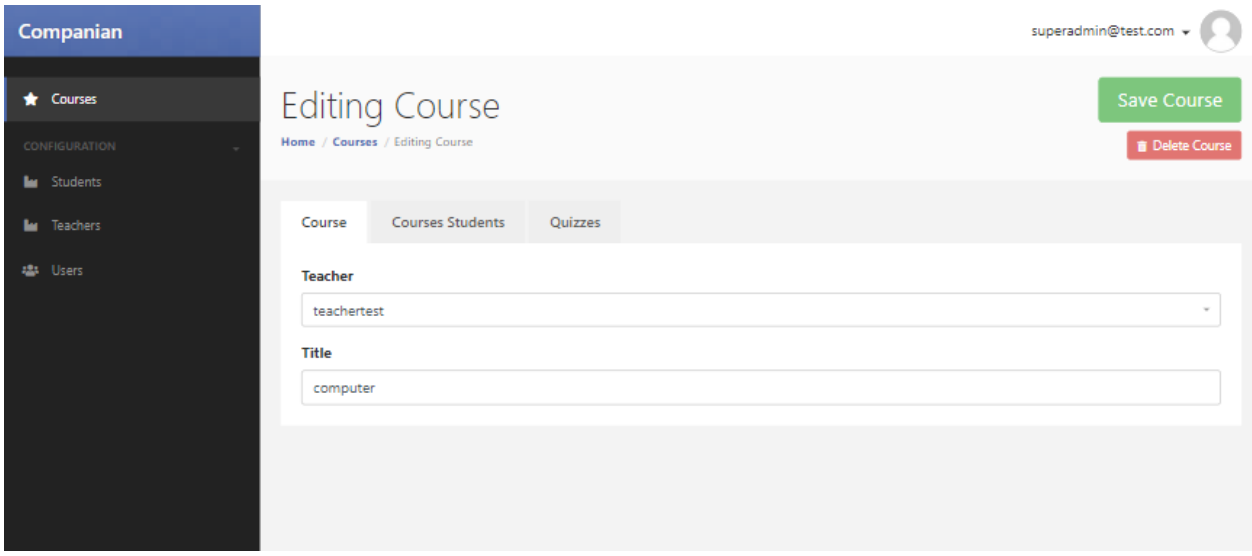
1 : def

2 : function

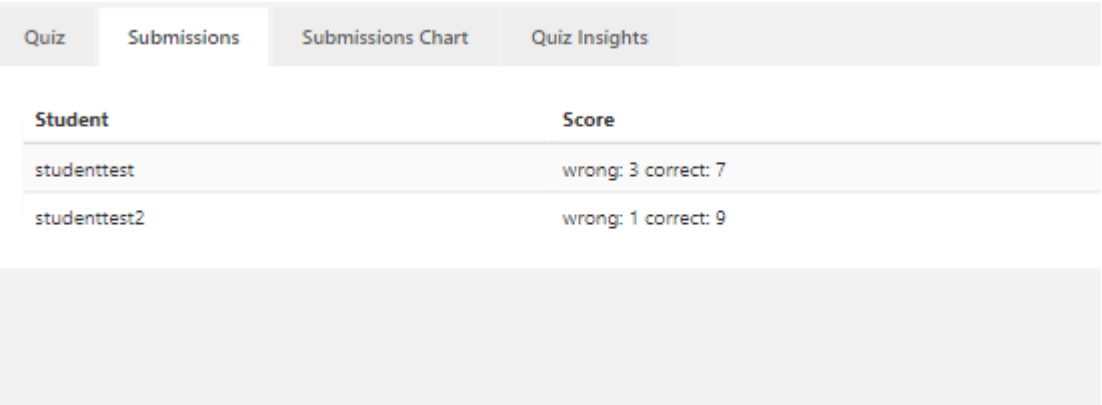
5.7.6. Teacher Record Portal



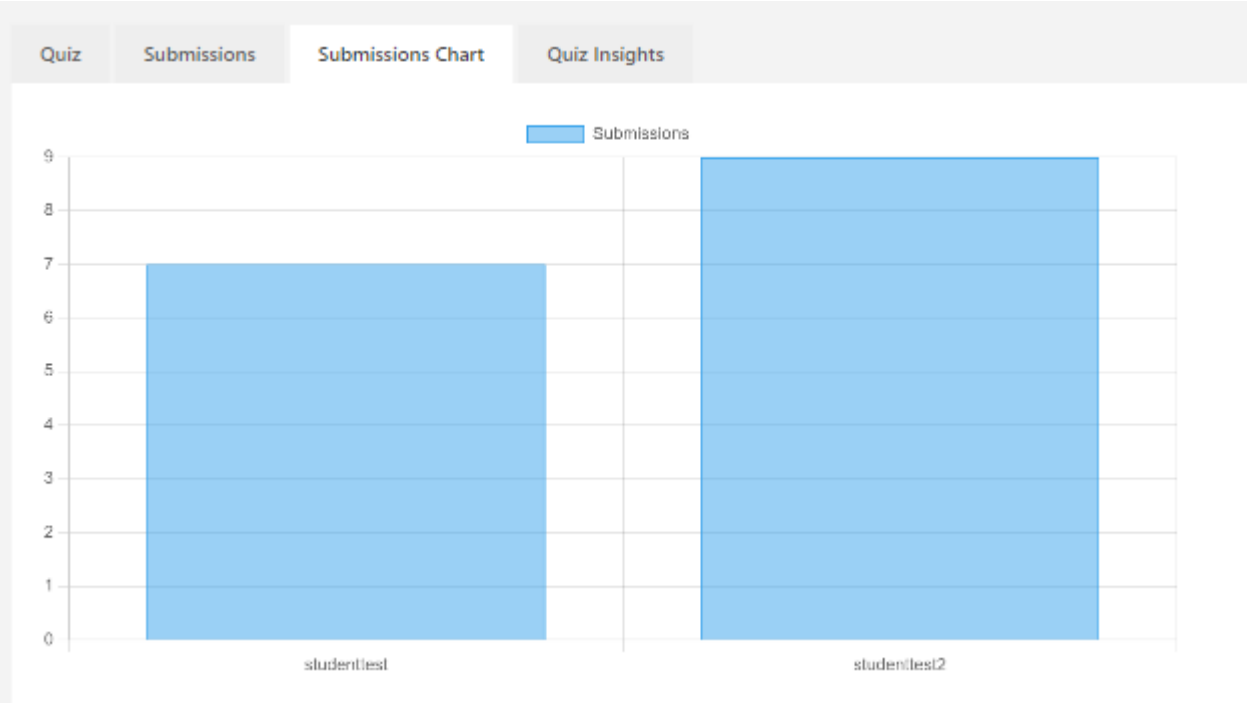
5.7.7. Editing Courses



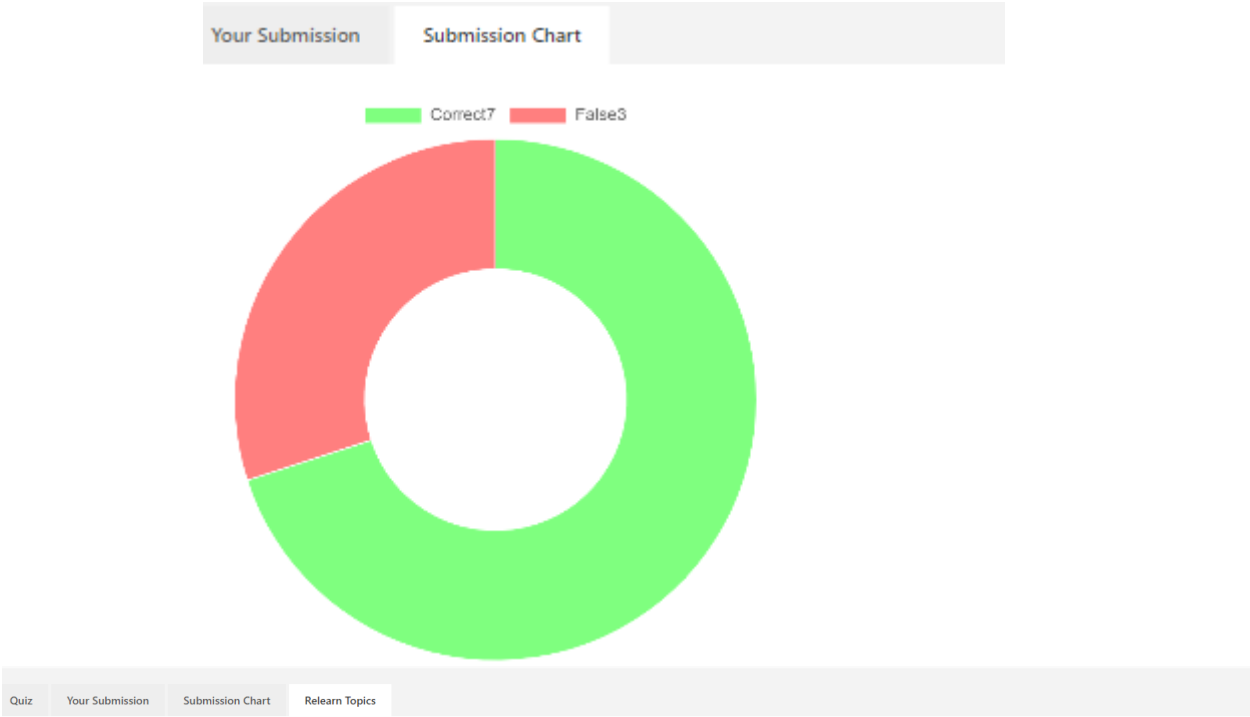
5.7.8. Submission Record

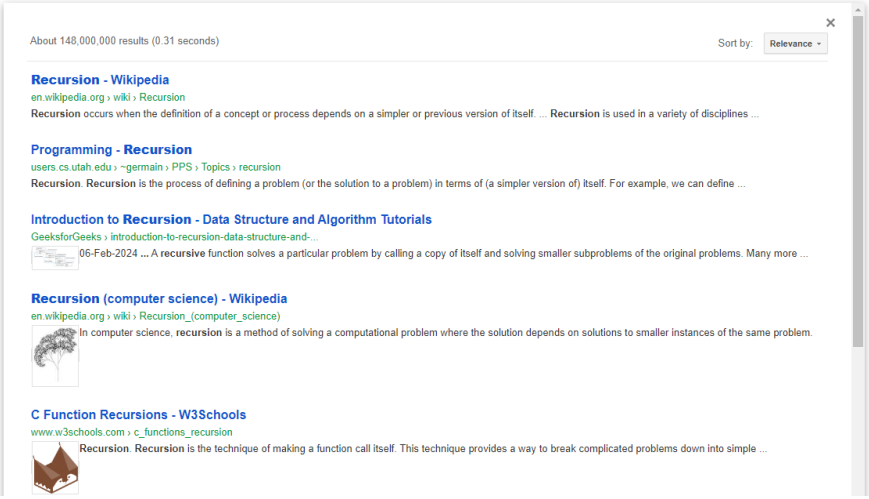


5.7.9. Student Submission Record



5.7.9. Quiz Insights


Your Submission
Submission Chart
Correct7
False3
Quiz
Your Submission
Submission Chart
Relearn Topics


About 148,000,000 results (0.31 seconds)
Sort by:
Relevance
Recursion - Wikipedia
en.wikipedia.org › wiki › Recursion
Recursion occurs when the definition of a concept or process depends on a simpler or previous version of itself. ... Recursion is used in a variety of disciplines ...
Programming - Recursion
users.cs.utah.edu › ~germain › PPS › Topics › recursion
Recursion. Recursion is the process of defining a problem (or the solution to a problem) in terms of (a simpler version of) itself. For example, we can define ...
Introduction to Recursion - Data Structure and Algorithm Tutorials
GeeksforGeeks › introduction-to-recursion-data-structure-and-...
06-Feb-2024 ... A recursive function solves a particular problem by calling a copy of itself and solving smaller subproblems of the original problems. Many more ...
Recursion (computer science) - Wikipedia
en.wikipedia.org › wiki › Recursion_(computer_science)
In computer science, recursion is a method of solving a computational problem where the solution depends on solutions to smaller instances of the same problem.
C Function Recursions - W3Schools
www.w3schools.com › c_functions_recursion
Recursion. Recursion is the technique of making a function call itself. This technique provides a way to break complicated problems down into simple ...

5.7.9. Result

Quiz | Your Submission | Submission Chart | Relearn Topics

1 What is recursion?

false

Recursion is a programming technique where a function calls itself in order to solve a problem.

2 What is the base case in recursion?

false

The base case is the condition in a recursive function that stops the recursion from continuing indefinitely.

3 What is the purpose of recursion?

false

Recursion is used to break down a complex problem into smaller, more manageable subproblems.

4 What is a recursive function?

false

A recursive function is a function that calls itself within its definition.

5 What happens if a recursive function does not have a base case?

true

6 What is the difference between direct and indirect recursion?

true

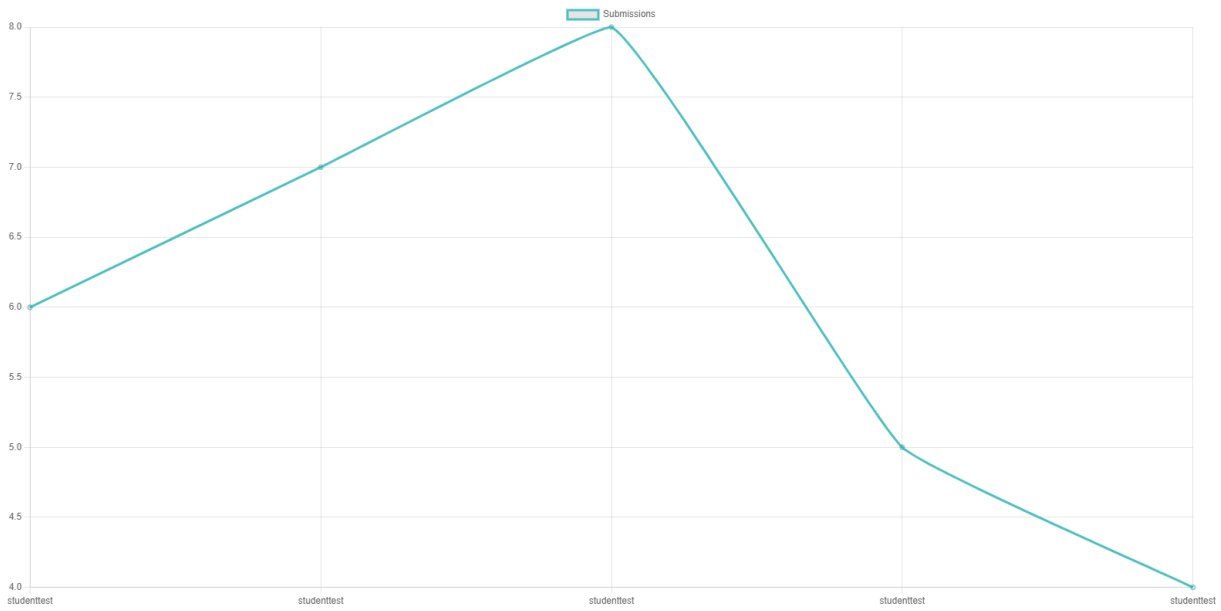
7 What is the role of the call stack in recursion?

false

The call stack is used to keep track of function calls in recursion, allowing the program to return to previous function calls when necessary.

8 What is tail recursion?

Course | Progress | Quizzes



Chapter 6

Testing and Evaluation

Chapter 6: Testing and Evaluation

Introduction:

Welcome to our exciting AI Classroom Project! We are building a special computer program that helps students learn and teachers teach using the latest technology. This program works on computers, phones, and tablets, so you can learn from anywhere!

What's in Our Program?

- **For Students:** You can join classes, watch fun videos, take quizzes that a smart computer helps make, and see how well you're doing in your subjects.
- **For Teachers:** You can create quizzes, see your students' progress, give grades, and manage your classes all in one place.

We are using some amazing tools like Laravel, OpenAI, React Native, and React to make our program really helpful and fun to use. This means it will be easy for you to learn new things and for teachers to help you even better.

Our goal is to make learning fun and easy for everyone. We want our program to be a place where you love to learn and feel excited about what you can achieve. So, let’s get ready to explore, learn, and grow together with our AI Classroom Project!

6.1. Use Case Testing

Use case testing is like practicing a school play before the big performance. We want to make sure everything works just right when it's showtime. Here’s how we do it for our learning program:

1. **Pick a Scenario:**
    - We think about different things that students and teachers might do with our program, like logging in, watching videos, or taking quizzes.
2. **Make a Plan:**

- We write down steps for each thing, like a recipe. For example, what happens first, what comes next, and what it looks like if it's working correctly.

3. **Act It Out:**

    - We pretend we are students and teachers using the program. We follow our plan and see if we can do everything we're supposed to do, like finishing a quiz or checking grades.

4. **Look for Mistakes:**

    - While we act it out, we watch closely to see if anything goes wrong or feels hard to do. Maybe a button doesn't work, or something is too hard to find.

5. **Fix Problems:**

    - If we find mistakes, we go back and fix them. It's like erasing a spelling error in your homework.

6. **Try Again:**

    - After we fix things, we test them again to make sure everything now works just right.

7. **Get Feedback:**

    - Sometimes, we let some real students and teachers try it too. They can tell us if they like it or if something still needs to be better.

By doing use case testing, we make sure our program helps you learn and doesn't make schoolwork harder. This testing helps us make everything just right for you and your teachers!

6.2. Equivalence partitioning

Equivalence partitioning is a way we test our learning program to make sure it works well for everyone. It's like sorting things into groups where each group has similar things. Here's how we do it:

1. **Make Groups:**
    - We think about all the different ways you might use our program, like entering answers in a quiz or choosing videos to watch.
    - Then, we divide these activities into groups based on what they have in common. For example, one group could be all the ways to answer a quiz question (like choosing a multiple-choice answer or typing a text).
2. **Choose Examples:**
    - From each group, we pick a few examples to test. We choose examples that are very typical for that group, so we don't have to test every single possibility. It's like choosing a couple of crayons from a box to see if the whole box is good.
3. **Test the Examples:**
    - We test these examples to make sure they work just right. If one example in a group works, we think that all the others in that group will work too. It's a bit like checking a few pieces of a puzzle to guess if the whole puzzle can be solved.
4. **Look for Mistakes:**
    - While testing, we look for any problems or mistakes. If we find something wrong in one example, we check all the other examples in the same group too.
5. **Fix and Check Again:**
    - If we find a problem, we fix it. Then we test again to make sure our fix works not just for that one example, but for all similar ones in the group.

Equivalence partitioning helps us make sure that every part of our program works well, whether it's watching videos, taking quizzes, or checking grades. It's like making sure every piece in a set of legos fits just right, so you can build amazing things without any trouble.

6.3. Boundary value analysis

Boundary value analysis is a special way we check our learning program to make sure it works perfectly, especially at the edges of what it's supposed to do. Think of it like testing the edges of a puzzle piece to make sure it fits just right with its neighbors. Here's how we do it:

1. **Find the Edges:**
    - We look for places in our program where you can put in numbers or make choices, like the highest or lowest number of questions you can choose for a quiz or the oldest and youngest ages that can use the app.
    - These edges are important because they help us make sure the program behaves nicely even when you use the extreme options.
2. **Test the Edges:**
    - We then test these edge points, like what happens if you try to enter zero questions for a quiz or the maximum number allowed. We also check what happens just inside these edges, like one question more or one question less.
    - It's like checking if a door can still open if it's a tiny bit bigger or smaller than the frame.
3. **Look for Mistakes:**
    - While testing, we watch for any problems, like errors showing up or the program crashing. If something doesn't work at the edge, it might not work in other places either.
4. **Fix Problems:**

- If we find anything wrong, we fix it. This makes sure that our program is strong and doesn't break down when you use it in different ways.

5. **Test Again:**

    - After fixing, we check those edges again to make sure everything now works smoothly. This is like double-checking that the puzzle piece fits perfectly after you've trimmed it a bit.

By doing boundary value analysis, we make sure our program is safe and fun to use, even if you try to use it in unusual ways. This helps us keep the learning adventure exciting and trouble-free for everyone!

6.4. Data flow testing

Data flow testing is like following the path of a secret message as it travels through a maze. We use it to check that all the information in our program goes to the right places without getting lost or mixed up. Here's how we do it:

1. **Map the Journey:**
    - First, we draw a map of how data (like your quiz answers or video choices) moves from one part of our program to another. Imagine drawing a path from the start to the end of a maze, showing where the message needs to go.
2. **Find Important Spots:**
    - We look for important spots along this path, like where your answers get checked or where your scores get calculated. These spots are like checkpoints in the maze where the message can change direction.
3. **Test Each Spot:**
    - We then test each of these spots to make sure that the data is handled correctly. For example, we check to make sure that when you answer a quiz question, the program correctly scores it and updates your grade.
    - It's like making sure that at every checkpoint in the maze, the message is read correctly and sent along the right path.
4. **Look for Mistakes:**

- While testing, we watch for any errors, like data getting lost (like a message that disappears in the maze) or going to the wrong place (like a message that takes a wrong turn).

5. **Fix Problems:**
    - If we find any issues, like a piece of data that doesn't go where it's supposed to, we fix it. This might mean changing the path a little or making the instructions clearer at each checkpoint.
6. **Test Again:**
    - After fixing things, we test the paths again to make sure everything works perfectly. It's like running the message through the maze again to make sure it can reach the end without any problems.

Data flow testing helps us ensure that our program works smoothly and that all the information you see and use is accurate and reliable. It keeps the learning adventure clear and fun for everyone!

6.5. Unit testing

In this project, unit testing is conducted by examining each piece or unit of code separately to confirm its correct operation. For instance, unit tests could be developed to check if the function tasked with creating quiz questions delivers the correct results when given various inputs. Additionally, tests might be set up to assess the correctness of the function that scores student responses. By implementing unit testing, it becomes feasible to spot and correct mistakes early in the development phase, which enhances the project's overall quality and dependability. Testing code units individually allows for a more effective bug detection and resolution process, making sure the project performs as expected.

6.6. Integration testing

Integration testing is like making sure all the pieces of a big puzzle fit together perfectly. When we build our learning program, we create lots of different parts, like video watching, quiz taking, and grade checking. Integration testing helps us check that all these parts work together smoothly. Here's how we do it:

1. **Put the Pieces Together:**

    - First, we take the different parts of our program that we have made and put them together. It's like when you start putting puzzle pieces together to see a bigger picture.

2. **Check the Connections:**

    - We look at how these parts connect and talk to each other. For example, we check if the part that plays videos can send you to the quiz part after you finish watching a video.

    - It's like making sure that each puzzle piece not only fits but also matches the picture it's supposed to create with the neighboring pieces.

3. **Test the Whole System:**

    - We then test the whole system to see if you can do things like log in, watch a video, take a quiz, and see your grades without any problems. We want to make sure that moving from one part of the program to another is easy and nothing stops working.

    - This is like checking if you can see the whole picture clearly once all the puzzle pieces are in place.

4. **Find and Fix Problems:**

    - If we find any part where the program doesn't work well or the parts don't connect right, we fix it. Maybe a button doesn't take you to the right next step, or maybe a score from a quiz doesn't show up in your grades right away.

    - It's like noticing that a piece of the puzzle is sticking up and not sitting flat, so you press it down or trim it a bit to make it fit better.

5. **Test Again:**

    - After we fix any issues, we test everything again to make sure that now it all works perfectly. We keep testing and fixing until everything is just right.

By doing integration testing, we make sure that our learning program is like a completed puzzle where every piece fits perfectly, so you have a great time learning without any hassle.

6.7. Performance testing

Performance testing is like seeing how fast a car can go and how well it can handle busy roads. We do this kind of testing for our learning program to make sure it works fast and smoothly, even when lots of students and teachers are using it at the same time. Here's how we do it:

1. **Test for Speed:**
    - We check how quickly our program can do things like open up a video or show your quiz results. It's important that you don't have to wait a long time for things to load, especially when you're eager to learn.
2. **Test with Many Users:**
    - We also see how our program handles it when lots of people try to use it at the same time. Imagine everyone in your school going to the cafeteria at once; we make sure the program can handle a big crowd without slowing down.
3. **Check for Bugs Under Pressure:**
    - Just like testing a car on different roads, we try our program in different situations to make sure it doesn't have any problems when it's very busy or when a lot of things are happening at once.
4. **Make Improvements:**
    - If we find that our program is slow or if it stops working when too many people use it, we work on fixing these problems. It's like tuning a car to make it run better.
5. **Test Again:**
    - After we make changes, we test everything again to make sure our program is now running faster and can handle lots of users at the same time without any issues.

Performance testing helps us ensure that our learning program is always ready and efficient, so your learning experience is smooth and enjoyable. It's like making sure the car is always ready for a fun road trip!

6.8. Stress Testing

Stress testing is like seeing how much weight a bridge can hold before it becomes unsafe. We do this kind of testing for our learning program to make sure it can handle a lot of stress, which means lots of students using it at the same time without breaking down. Here's how we do it:

1. **Push the Limits:**
    - We try to push our program to work harder than usual. For example, we see what happens when hundreds of students try to watch videos or take quizzes at the same time. It's like seeing how many books you can pile on a shelf before it gets too heavy.
2. **Look for Weak Spots:**
    - While we're pushing the program, we watch to see where problems pop up. Maybe the website starts to load really slowly, or it stops working if too many people log in at once. It's like finding the weak spots in the bridge where it might start to crack.
3. **Fix the Problems:**
    - Once we know where the weak spots are, we try to fix them. This might mean making some parts of the program stronger or faster. It's like reinforcing parts of the bridge to make it hold more weight.
4. **Test Again:**
    - After we make changes, we do the stress test again to make sure that our fixes worked. We keep doing this until the program can handle the biggest crowds without any trouble.
5. **Ensure Safety and Comfort:**
    - Our goal is to make sure that no matter how many students are using the program at once, it stays fast and easy to use. It's like making sure the bridge is safe and strong enough for everyone to cross at the same time.

Stress testing helps us ensure that our learning program won't let you down, even on the busiest school days. It's all about making sure you have a reliable and smooth experience every time you use it!

Chapter 7

Summary, Conclusion, and Future Enhancements

Chapter 7: Summary, Conclusion & Future Enhancements

7.1. Project Summary

We've created a cool learning program that works on the internet and on mobile phones, so students and teachers can use it anywhere! Here's what our project does:

For Students:

- **Join Classes:** You can see all the classes you're in and join new ones.
- **Watch Videos:** There are lots of videos to help you learn things in a fun way.
- **Take Quizzes:** You can take quizzes that our smart AI makes just for you, to see how much you've learned.
- **Check Grades:** After you take quizzes, you can look at your grades to see how well you did.

For Teachers:

- **Create Quizzes:** Teachers can make quizzes with the help of AI, which makes sure the quizzes are good and helpful.
- **See and Give Grades:** Teachers can look at how students are doing and give them grades.
- **Manage Classes:** Teachers can create new classes or take care of the classes they already have.

We built this program using some awesome tools like Laravel, OpenAI, React Native, and React. These tools help make sure everything works smoothly and looks good.

Our Goals:

- We want to make learning fun and easy for everyone.

- We aim to help teachers do their jobs better with technology.
- We hope to make our program work well on any device, so everyone can use it without any problems.

What We Did:

- We planned and built the program.
- We tested it to make sure it works right.
- We fixed any problems we found.

Now, our program is ready for schools to use, and we're excited to see how it helps students and teachers learn and teach better!

7.2. Achievements and Improvements

What We Achieved:

- **Building a Complete Learning Program:** We created a program that students and teachers can use to make learning and teaching easier and more fun. It works on the web and on mobile devices, so you can use it anywhere, anytime!
- **Using AI:** Our program uses AI to help make quizzes and suggest the best videos and materials for each student. This makes learning really personalized, which means it's just right for what you need.
- **Helping Teachers:** Teachers have new tools to make quizzes quickly, check student progress, and manage their classes more easily. This saves them time and lets them focus on teaching.

How We Improved:

- **Making It Faster:** We found that sometimes our program was a bit slow when lots of people were using it. So, we made improvements to make it faster, which means no more waiting around for things to load.
- **Fixing Bugs:** Like any new program, we had a few small glitches at first, such as buttons that didn’t work right or pages that loaded slowly. We fixed all these issues to make sure everything works smoothly.

- **Adding Features:** Based on feedback from students and teachers who tried our program, we added more cool features. For example, we put in a system where students can earn points for good work, which makes learning more like a game.

Next Steps:

- **Keep Making It Better:** We're not done yet! We plan to keep listening to what students and teachers say about our program so we can keep making it better.
- **Add More Subjects:** Right now, we have lots of great materials for some subjects, but we want to add even more. We plan to include more subjects so everyone can use our program for their different classes.

We're really proud of what we've accomplished, and we're excited to keep improving our program. Our goal is to make a learning platform that makes everyone love learning and makes school a little easier and a lot more fun!

7.3. Critical Review

What We Did Well:

- **Fun and Easy to Use:** We made a program that's not just helpful for learning, but also fun! Students enjoy using it because it feels like playing a game while learning.
- **Works on Many Devices:** Our program can be used on computers, tablets, and smartphones. This is great because it means you can learn from anywhere, whether you're at home or on the go.
- **Smart Learning with AI:** The AI in our program helps make learning better for each student. It suggests videos and quizzes that are just right for what you need to learn next.

What Could Be Better:

- **Needs More Subjects:** Right now, our program has lots of materials for subjects like math and science, but we could add more for subjects like history and art. Some students and teachers have said they want more variety.
- **Sometimes Slow:** When lots of people use the program at the same time, it can get a bit slow. We've made it faster than it was, but we need to keep working on this to make sure it's always quick and responsive.

- **More Teacher Tools:** Some teachers have told us they would like more tools for checking how much students are learning. We could improve the program by adding more features that help teachers track student progress over time.

What We Learned:

- **Feedback is Important:** Listening to what students and teachers say about our program is really helpful. It tells us what's working and what's not, so we can make it better.
- **Testing is Key:** We found some bugs when we first launched the program because we didn't test it enough in different situations. We've learned that we need to do more testing before we say something is finished.

Moving Forward:

- We're planning to add more subjects and more tools for teachers. We also want to make sure our program stays fast and fun to use, no matter how many students are online. We're excited to keep improving and making our program the best it can be!

7.4. Lessons Learnt

When we created our learning program, we learned a lot of important things that will help us do even better in the future. Here are some of the big lessons:

1. **Testing is Super Important:**

    - We learned that we need to test our program in lots of different ways before we share it with everyone. Just like when you practice a lot for a school play to make sure everything goes right on the big day, we need to make sure our program works perfectly in all kinds of situations.

2. **Listen to Users:**

    - Feedback from students and teachers is really valuable. They use our program every day, so they know best what works well and what doesn't. Listening to their ideas helps us make the program better and more fun for everyone.

3. **Plan for Changes:**

    - Sometimes things don't go as planned, and that's okay! We learned that we need to be ready to make changes and fix problems quickly. It's like when you're building a lego set and find out you put a piece in the wrong place; you need to be ready to fix it and keep going.

4. **Make It Fun:**

    - We found out that making learning fun is super important. When students enjoy using our program, they learn more and do better in school. We want to keep adding fun features, like games and rewards, to keep students excited about learning.

5. **Keep Improving:**

    - Technology changes fast, and we need to keep updating our program so it stays modern and useful. It's like updating your game console to play the newest games; we need to keep our program up-to-date so students and teachers always have the best tools.

6. **Work Together:**

    - Working as a team and sharing ideas made our project better. We learned that when everyone helps and offers their skills, we can create something really great.

These lessons will help us as we continue to improve our program and start new projects. We're excited to use what we've learned to make learning even better for students everywhere!

7.5. Future Enhancements/Recommendations

As we keep working on our learning program, we have some cool ideas to make it even better! Here are the things we want to do:

1. **Add More Subjects:**
    - We plan to include more lessons and quizzes for different subjects like history, art, and music. This way, everyone can find something they love to learn about in our program.
2. **Make It Faster:**
    - We want to make sure our program is super fast, even when lots of students are using it at the same time. We'll work on making the videos and quizzes load quicker, so you don't have to wait.
3. **Improve AI Features:**
    - Our smart AI can help make learning just right for you. We're going to make it even smarter, so it can suggest the perfect videos and quizzes based on what you like and what you need to learn more about.
4. **Create More Games:**
    - Learning is more fun when it feels like playing. We'll add more educational games that help you learn important things while you're having fun.
5. **Enhance Teacher Tools:**
    - We'll give teachers more tools to help them see how each student is doing. This will make it easier for teachers to help students who need it and to know when everyone is doing great.
6. **Make the Program Easier to Use:**
    - We want everyone to be able to use our program easily, so we'll keep making the design simpler and more friendly. We'll also make sure it works great on all devices, like tablets and smartphones.
7. **Focus on Security:**
    - Keeping your information safe is super important. We're going to make sure that all your work and grades are secure, so only you and your teachers can see them.
8. **Listen to Feedback:**
    - We'll keep asking for ideas and suggestions from students and teachers and use that feedback to make our program better.

We're really excited about making these enhancements! They'll help make our learning program a favorite tool for students and teachers everywhere. We can't wait to see how it helps everyone learn better and have more fun doing it!

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AI based learning

ORIGINALITY REPORT

9%	8%	0%	9%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	Submitted to Higher Education Commission Pakistan Student Paper	7%
2	www.coursehero.com Internet Source	1%
3	Submitted to Universidad TecMilenio Student Paper	<1%
4	Submitted to Washington State University System Student Paper	<1%
5	Submitted to Melbourne Institute of Technology Student Paper	<1%
6	Submitted to National University of Singapore Student Paper	<1%
7	Submitted to Asia Pacific University College of Technology and Innovation (UCTI) Student Paper	<1%